Ellipse Perimeter: Ramanujan Approximation and N-Series Geometric Acceleration

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Abstract

Ramanujan's formulas for π exhibit extraordinary acceleration through modular and hypergeometric structure. The N-Series framework approaches the same constant from a different direction: it begins with constructible polygonal approximations to the circle and accelerates them by cancelling successive even-power error terms. This note compares these two viewpoints. Ramanujan-type formulas reveal a deep analytic acceleration mechanism, while the N-Series construction defines a geometric limiting constant, denoted π_g, through chordal polygonal refinement. The comparison is complementary rather than competitive: Ramanujan's formulas give exceptional closed-form series acceleration, while the N-Series gives a constructive geometric cancellation pathway to π.

1 Exact Ellipse Perimeter

For an ellipse with semi-major axis a and semi-minor axis b, the exact perimeter is expressed in terms of the complete elliptic integral of the second kind:

$$P_{\text{exact}} = 4a\,E(e), \qquad e = \sqrt{1 - \frac{b^2}{a^2}},$$

where

$$E(e) = \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t}\,\mathrm{d}t.$$

This exact expression is not elementary in general. For this reason, the ellipse perimeter has long served as a classical testing ground for geometric, algebraic, and numerical approximation methods.

2 Ramanujan's Approximation

Ramanujan gave several remarkably accurate closed-form approximations for the ellipse perimeter. One of the most widely used is

$$P_{\text{R}} = \pi(a+b)\left[1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right], \qquad h = \left(\frac{a-b}{a+b}\right)^2.$$

This formula is exceptional because it compresses elliptic-integral behavior into a simple algebraic expression involving a, b, and h. Its strength is that it gives a high-accuracy perimeter estimate without requiring an explicit refinement process.

3 Polygonal Approximation and N-Series Acceleration

The N-Series viewpoint approaches the same problem from the opposite direction. Instead of seeking a single closed-form approximation, one begins with a geometric polygonal approximation to the ellipse and then accelerates the convergence by cancelling leading error terms.

Let P_N denote the perimeter of the inscribed N-segment polygon obtained by sampling the ellipse

$$x(t) = a\cos t, \qquad y(t) = b\sin t, \qquad 0 \le t < 2\pi,$$

at equally spaced parameter values

$$t_j = \frac{2\pi j}{N}, \qquad j = 0, 1, \ldots, N-1.$$

Then

$$P_N = \sum_{j=0}^{N-1} \sqrt{\big(x(t_{j+1}) - x(t_j)\big)^2 + \big(y(t_{j+1}) - y(t_j)\big)^2},$$

with $t_N = t_0 + 2\pi$.

For smooth closed curves, such symmetric polygonal approximations possess an even-power asymptotic expansion of the form

$$P_N = P_{\text{exact}} + \frac{C_2}{N^2} + \frac{C_4}{N^4} + \frac{C_6}{N^6} + \cdots .$$

This is precisely the structure targeted by the N-Series cancellation operators.

The two-scale accelerated perimeter is

$$\widehat{P}_N^{(2)} = P_{2N} + \frac{P_{2N} - P_N}{2^2 - 1}.$$

If the leading error term is $C_2 N^{-2}$, this operator cancels it exactly and gives

$$\widehat{P}_N^{(2)} = P_{\text{exact}} + O(N^{-4}).$$

Using the three scales N, $2N$, and $4N$, one may cancel the N^{-2} and N^{-4} terms simultaneously. The resulting three-scale operator is

$$\widehat{P}_N^{(3)} = \frac{1}{45}\,P_N - \frac{4}{9}\,P_{2N} + \frac{64}{45}\,P_{4N}.$$

(Its three weights sum to 1, and they are fixed by the requirement that the N^{-2} and N^{-4} terms vanish; one checks $\frac{1}{45} - \frac{1}{4}\cdot\frac{4}{9} + \frac{1}{16}\cdot\frac{64}{45} = 0$ and the analogous N^{-4} condition.) This gives

$$\widehat{P}_N^{(3)} = P_{\text{exact}} + O(N^{-6}),$$

provided the even-power expansion holds.

4 Numerical Comparison

We tested ellipses with $a = 1$ and

$$b \in \{0.90,\ 0.75,\ 0.50,\ 0.25\}.$$

For each ellipse, the exact perimeter was computed from the elliptic-integral formula and compared against Ramanujan's approximation, the raw polygonal perimeter P_N, the two-scale N-Series approximation of order $O(N^{-4})$, and the three-scale N-Series approximation of order $O(N^{-6})$.

The results show two complementary regimes. For the near-circular ellipse $a = 1$, $b = 0.90$, Ramanujan's formula is exceptionally accurate, with absolute error approximately 2.24×10^{-17}. At $N = 256$, the three-scale N-Series approximation reaches error approximately 6.32×10^{-17}, which is comparable but does not yet improve on Ramanujan's value.

For more eccentric ellipses, however, the N-Series ladder becomes dominant. At $a = 1$, $b = 0.75$, Ramanujan's error is approximately 4.61×10^{-13}. The three-scale N-Series approximation improves on this already at $N = 64$, where the error is approximately 2.40×10^{-13}, and by $N = 256$ reduces the error to approximately 5.85×10^{-17}.

The improvement is more pronounced for $a = 1$, $b = 0.50$. Here Ramanujan's error is approximately 2.21×10^{-9}, whereas the three-scale N-Series approximation at $N = 32$ already gives error approximately 7.34×10^{-12}. By $N = 256$, the three-scale N-Series error is approximately 5.13×10^{-17}.

For the more eccentric case $a = 1$, $b = 0.25$, Ramanujan's error is approximately 1.07×10^{-6}. The three-scale N-Series approximation improves on this even at $N = 32$, with error approximately 1.81×10^{-7}, and reaches approximately 4.54×10^{-17} by $N = 256$.

The three-scale error approximately falls by a factor of 64 per scale doubling — for $b = 0.50$, the successive values are 2.10×10^{-13} at $N = 64$, 3.28×10^{-15} at $N = 128$, and 5.13×10^{-17} at $N = 256$ — in agreement with the predicted $O(N^{-6})$ order.

a	b	N	Ramanujan error	Three-scale N-Series error
1	0.90	256	2.24×10^{-17}	6.32×10^{-17}
1	0.75	256	4.61×10^{-13}	5.85×10^{-17}
1	0.50	256	2.21×10^{-9}	5.13×10^{-17}
1	0.25	256	1.07×10^{-6}	4.54×10^{-17}

Table 1: Comparison of Ramanujan's closed-form ellipse perimeter approximation with the three-scale N-Series accelerated polygonal approximation.

5 Interpretation

The comparison does not show a single universal winner. Rather, it shows two different acceleration mechanisms.

Ramanujan's formula gives an exceptional one-shot algebraic approximation, especially near the circular case. The N-Series method gives a systematic geometric refinement ladder whose advantage increases as the ellipse becomes more eccentric. In this sense, the N-Series does not replace Ramanujan's approximation; it complements it by supplying a constructible, scale-cancellation pathway for driving the polygonal ellipse perimeter to high precision.

This distinction is important. Ramanujan's approximation is closed-form and requires no refinement parameter N. The N-Series approximation is algorithmic: it begins with chordal polygon

approximations and then cancels the leading even-power error terms. Thus the two methods illuminate different structures in the same classical problem. Ramanujan reveals a hidden algebraic approximation to the elliptic integral, while the N-Series reveals a scale-cancellation structure inside the polygonal geometry of the ellipse.

6 Conclusion

The ellipse perimeter problem provides a useful bridge between Ramanujan-type approximation and N-Series geometric acceleration. Ramanujan's approximation is extraordinarily effective near the circular case and remains impressive even for significant eccentricity. The N-Series refinement ladder, however, systematically improves with scale and overtakes the Ramanujan approximation for the tested moderate and high eccentricities.

The conclusion is therefore complementary rather than competitive: Ramanujan's formula supplies a powerful closed-form algebraic estimate, while the N-Series supplies a systematic geometric pathway to high precision through exact cancellation of even-power polygonal error terms.

References

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